

# Foundations of Data Science for Biologists

## NumPy

BIO 724D

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# Learning objectives for today

Become familiar with NumPy's **ndarray** data structure

Learn how to **create**, **modify**, **index**, and **query** ndarrays

Apply these skills to manipulating and analyzing images

# Introduction to NumPy and ndarrays



**NumPy** is a foundational library for scientific computing with Python

Supports intensive computation with arrays

- Highly efficient multidimensional array data class

- High-performance functions and methods

Foundational: used by many other libraries

- Algebra, statistics, discrete math (e.g., SciPy, Clifford, SymPy)

- Data science (e.g., Pandas, GeoPandas)

- Artificial intelligence, machine learning, deep learning (e.g., TensorFlow, PyTorch)

- Image processing, signal processing (e.g., SciPy, BioSPPy)

- Image/video processing, simulations, modeling (e.g., Scikit-image, Pillow)



# ndarrays

**ndarray** (n-dimensional array) is the core data structure in NumPy

Stores **homogenous data** only

Most common: integers or floats (primary intended use case)

Also possible: Booleans, strings, lists, sets, tuples, etc. (not optimized for these)

Allows for data objects in any number of dimensions

**1D** (vector), **2D** (array), **3D** (matrix), **4D** (tensor), and more

Designed for **functional programming**

Most operations are fully vectorized, including plain Python operators and functions

